Haiwell CBOX-E ↔ RHEL IIoT Setup

Configuration reference | Prepared by Nipun | 05 Oct 2026

1. Network

Device	Interface	IP address	Purpose
CBOX-E	LAN1	`192.168.1.112`	Edge gateway
RHEL PC	enp2s0 (built-in LAN)	`192.168.1.50` + `192.168.1.200`	CBOX network + MQTT broker IP
RHEL PC	enp0s20u1 (USB adapter)	192.168.229.x (DHCP)	Internet
Windows laptop	Ethernet	`192.168.1.60`	Haiwell SCADA config / download
Windows laptop	Wi-Fi	10.30.5.19	Office network
RHEL NM profile	enp2s0	`cbox`	Static IP connection profile

2. CBOX Device

Item	Value
Machine code	`70718373000501130 11`
OS / HW / APP version	v1.1.1 / CBOX-CPU-V1.2.0 / 3.50.1.13
Settings page	`http://192.168.1.112/setting`
Project page	`http://192.168.1.112/project/apps/index`
Project	RHEL_IoT_Task (model CBOX-7, 800x480)
Project file (Windows)	`C:\Users\nipun\Desktop\RHEL_IoT_Task.hwdev`
Runtime file	`C:\Users\nipun\Desktop\Runtime\RHEL_IoT_Task.hwrun`
Config software	Haiwell Cloud SCADA 3.50.1.19
Download password	None

3. Haiwell MQTT Device (MQTT_1)

Item	Value
Device type / channel	MQTT / TCP
Broker	`192.168.1.200:1883`
Username / password	None (to be added - see section 9)
Data group	data1, identifier `factory/task1/`, QoS 0 (at most once)
Terminal	CBOX, code 1
Channel	test_var -> id V0, Integer, length 1, Read & write
Variable name	`data1_CBOX_test_var`
Screen element	NumShowIput_1 (Numeric input/display on Main_display)

4. MQTT Message Format

Direction	Topic	Payload
RHEL -> CBOX	`data/1234/factory/task1//1`	`{"V0":"123","_groupName":"data1"}` (value as string)
CBOX -> RHEL	`cmd/1234/factory/task1//1`	`{"type":"set_var","payload":{"V0":"55"}}`
CBOX subscribes	`+/1234/factory/task1//1`	-
CBOX client ID	`70718373000501130 11/1`	-

5. Docker Containers (RHEL)

Container	Image	Port	Status / note
mosquitto	eclipse-mosquitto:2	1883	Running - MQTT broker (new)
cbox-postgres	postgres:16	5432	Running - database (new)
mqtt-bridge	bridge-app	-	Pre-existing; fails resolving host 'mqtt'
mysql-db	mysql:8.0	-	Pre-existing (flowmeter app)
flowmeter_web-app	web-app	-	Pre-existing

6. Mosquitto Broker

Item	Value
Config file	`~/mosquitto/config/mosquitto.conf`
Data folder	`~/mosquitto/data`
Listener	1883 on 0.0.0.0
Authentication	allow_anonymous true (to be secured)
Firewall	1883/tcp open (permanent)

7. PostgreSQL

Item	Value
Host / port	localhost:5432
Database	cbox_db
User	cbox
Password	Stored only in ~/cbox/cbox.env (not recorded here)
Table	cbox_tag_log (id, ts, tag_name, value, direction, raw)
Data folder	`~/pgdata`
Credentials file	`~/cbox/cbox.env` (chmod 600)

8. Python Bridge

Item	Value
Libraries	paho-mqtt, psycopg2-binary
Test script	`~/cbox_bridge.py` (MQTT only)
Main script	`~/cbox_bridge_db.py` (MQTT + PostgreSQL logging)
Run (current)	`python3 ~/cbox_bridge_db.py` (manual)

9. Remaining Tasks

#	Task	Status
1	PostgreSQL password + .env credentials file	In progress
2	Mosquitto username/password (update CBOX MQTT_1 + download)	Pending
3	Run Python bridge as a systemd service (auto-start)	Pending
4	Decide on mqtt-bridge container (fix alias or stop)	Pending - confirm with supervisor